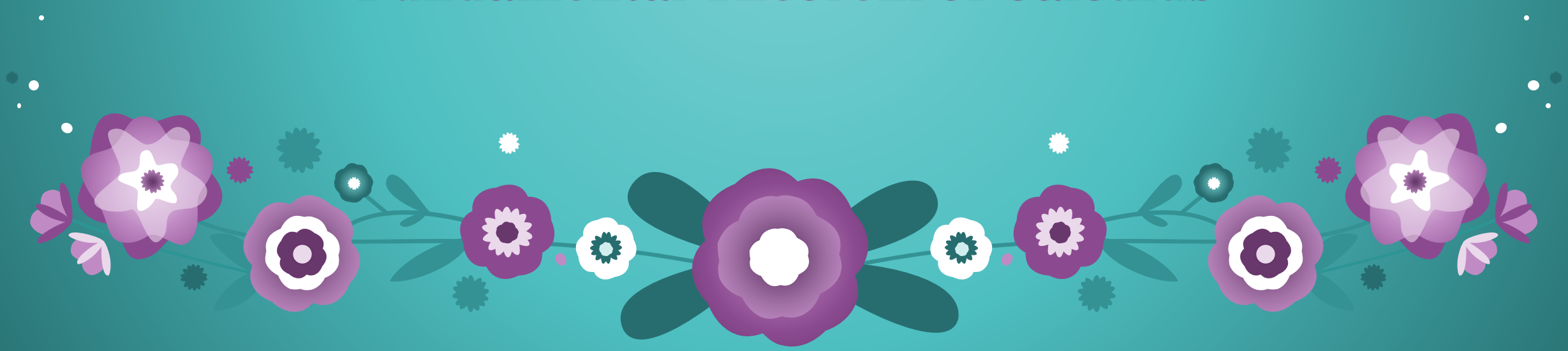
5.3

Fundamental Theorem of Calculus



What are the derivatives?

- Consider the relation between these functions
- F(x)= $x^3 - 5x + 2$
- $f(x) = 3x^2 - 5$
- F'(x)=f(x)

Then f is the derivative of F and F is an antiderivative of f

Remark

- A given function has more than one antiderivative as for $f(x) = 3x^2 - 5$

have the antiderivatives

F(x)= $x^3 - 5x + 2$

F(x)= $x^3 - 5x + 10$

F(x)= $x^3 - 5x + 9$

Basically F(x)= $x^3 - 5x + c$

Are all antiderivative as for $f(x) = 3x^2 - 5$

Definition

F(x)= is the derivative of f(x) if

$F'(x)=f(x)$

Example

- Find the antiderivative of f(x)=10x+4

$$\int (10x+4)\,dx$$

$$= \int (10x)\,dx + \int (4)\,dx$$

$$= 10\int (x)\,dx + \int (4)\,dx$$

↑ golden rule

$$= \frac{10x^2}{2} + 4x + C$$

$$= 5x^2 + 4x + C$$

Solution

- F(x)= $5x^2 + 4x$ +c

Fundamental Theorem of Calculus part I

If f is continuous on $[a, b]$, then the function F defined by

$$F(x) = \int_a^x f(t)dt$$

x ← variable

a ← number

must be in this order

is continuous on $[a, b]$ and differentiable on (a, b), and

- $\frac{d}{dx}F(x) = \frac{d}{dx}\int_a^x f(t)dt = f(x)$

will always match

Remark:

- Note that in FTC1 the lower limit of the integration is a number and the upper limit is x.
- Every continuous function f has an antiderivative F(x)
- This theorem connect the integration with differentiation.

Example

Find the derivative of

$$F(x)=\int_0^x (t^2-1)^{20}\,dt$$

$$f'(x)=\frac{d}{dx}\int_0^x (t^2-1)^{20}\,dt$$

$$f'(x)=(x^2-1)^{20}$$

Example

- Find the derivative of
- $F(x)= \int_x^2 (\cos(t)^2 + t)dt$

Switched then what its supposed to

$f(x) = -\int_2^x (\cos(t)^2 + t)\ dt$

$f'(x) = -\frac{d}{dx}\left[\int_2^x (\cos(t)^2 + t)\ dt\right]$

$f'(x) = -(\cos x^2 + x)$

Example:

- Find the derivative of
- $F(x)=\int_1^{\sqrt{x}} \frac{s^2}{s^2+1}\,ds$

$f'(x) = \frac{d}{dx}\int_1^{\sqrt{x}} \frac{s^2}{s^2+1}\,ds$

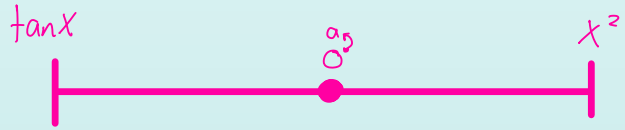
$= \frac{(\sqrt{x})^2}{(\sqrt{x})^2+1} \cdot (\sqrt{x})^2$ ← Math 111 Rule $\frac{1}{2\sqrt{x}}$

$= \frac{x}{x+1} \cdot \frac{1}{2\sqrt{x}}$

Example

Find the derivative of

$$g(x)=\int_{\tan x}^{x^2}\frac{1}{\sqrt{2+t^4}}\,ds$$



① $H(x)\int_{\tan x}^{0}\frac{1}{\sqrt{2+t^4}}dt+\int_{0}^{x^2}\frac{1}{\sqrt{2+t^4}}dt$ we split it because the gave us on the ∫ 2 variabels & no number

flip and add a negative ② $=-\int_{0}^{\tan x}\frac{1}{\sqrt{2+t^4}}dt+\int_{0}^{x^2}\frac{1}{\sqrt{2+t^4}}dt$ it must be Variable on top & number on the bottom

③ $H'(x)=\frac{d}{dx}\left[-\int_{0}^{\tan x}\frac{1}{\sqrt{2+t^4}}dt\right]+\frac{d}{dx}\int_{0}^{x^2}\frac{1}{\sqrt{2+t^4}}dt$ multiply it by its own inverse

④ $=\frac{-1}{\sqrt{2+(\tan x)^4}}(\tan x)'+\frac{1}{\sqrt{2+(x^2)^4}}(x^2)'$ find the inverse

⑤ $=\frac{-1}{\sqrt{2+(\tan x)^4}}\sec^2 x+\frac{1}{\sqrt{2+(x^2)^4}}(2x)$ ⑥ $=\frac{\sec^2 x}{\sqrt{2+(\tan x)^4}}+\frac{2x}{\sqrt{2+x^8}}$

Find the derivative of

$g(x)=\int_{x^2}^{x^3} \sqrt{t^4-8}\, ds$

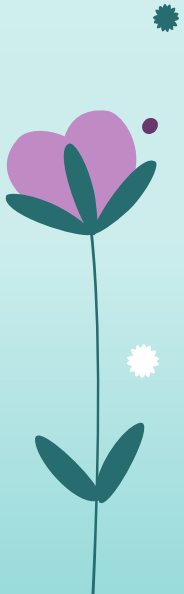
Example

Fundamental Theorem of Calculus part II

If f is continuous on $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

where F is any antiderivative of f,
that is, a function such that $F' = f$.



Example

Evaluate
$\int_2^5 [3x - 4]dx$

5.4

Evaluate

$$\int_1^4 [2x-3]^2 dx$$

1. Remove the square by using

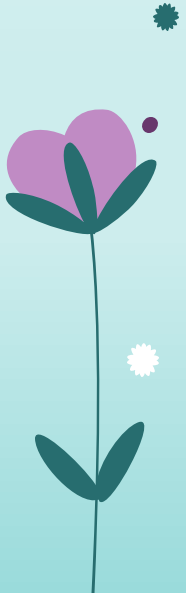
1) $\int_1^4 (4x^2 - 2(3)(2x) + 9)\,dx$

2) $\int_1^4 (4x^2)\,dx + \int_1^4 (12x)\,dx + \int (9)\,dx$

3) $\left[\frac{4x^3}{3} - \frac{12x^2}{2} + 9x + c\right]$

Golden rule

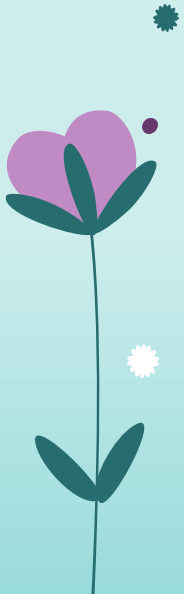
Example



Evaluate

$$\int_0^1 \frac{x+\sqrt{x}}{4}\,dx$$

Example



$$= -\left[2\frac{x^2}{x} - x + C_1\right]_0^{1/2} + \left[2\frac{x^2}{x} - x + C_2\right]_{1/2}^{0}$$

$$= \left[\left[(1/2)^2 - (1/2)\right] - \left[0^2 - 0\right]\right] - \left[\left[2^2 - 2\right] - \left[\frac{1}{2}^2 - \frac{1}{2}\right]\right]$$

find $\int (x^e + e^x)\,dx$

seperate

$= \int (x^e)\,dx + \int (e^x)\,dx$

$= \frac{x^{e+1}}{e+1} + e^x + c$

Ex: $\int \frac{2e^x}{\sinh x + \cosh x}\,dx$

$\cosh x = \frac{e^x + e^{-x}}{2}$

$\sinh x = \frac{e^x + e^{-x}}{2}$

$\frac{e^x + e^{-x}}{2} + \frac{e^x + e^{-x}}{2} = \frac{2e^x}{2} \to e^x$

$\int \frac{2e^x}{e^x}\,dx \to \int 2\,dx = 2x + c$

① for $f(x) = \int_a^x f(t)\,dt$

$f'(x) = \frac{d}{dx}\int_a^x f(t)\,dt \Rightarrow f(x)$

② for $H(x) = \int_a^{g(x)} f(t)\,dt$

$H'(x) = \frac{d}{dx}\int f(t)\,dt \Rightarrow f(g(x)) \cdot g'(x)$

find $H(x) = \int_{\cos x}^{\sin x} \ln(1+2t)\,dt$

find $H'(x)$

$\cos x$ ——— 0 ——— $\sin x$

$= \int_{\cos x}^{0} \ln(1+2t)\,dt + \int_0^{\sin x} \ln(1+2t)\,dt$

$= -\frac{d}{dx}\int_0^{\cos x} \ln(1+2t)\,dt + \frac{d}{dx}\int_0^{\sin x} \ln(1+2t)\,dt$

$= -\ln(1+2\cos x)(\cos x)' + \ln(1+2\sin x)(\sin x)'$

$= +\ln(1+2\cos x)(+\sin x) + \ln(1+2\sin x)(\cos x)$